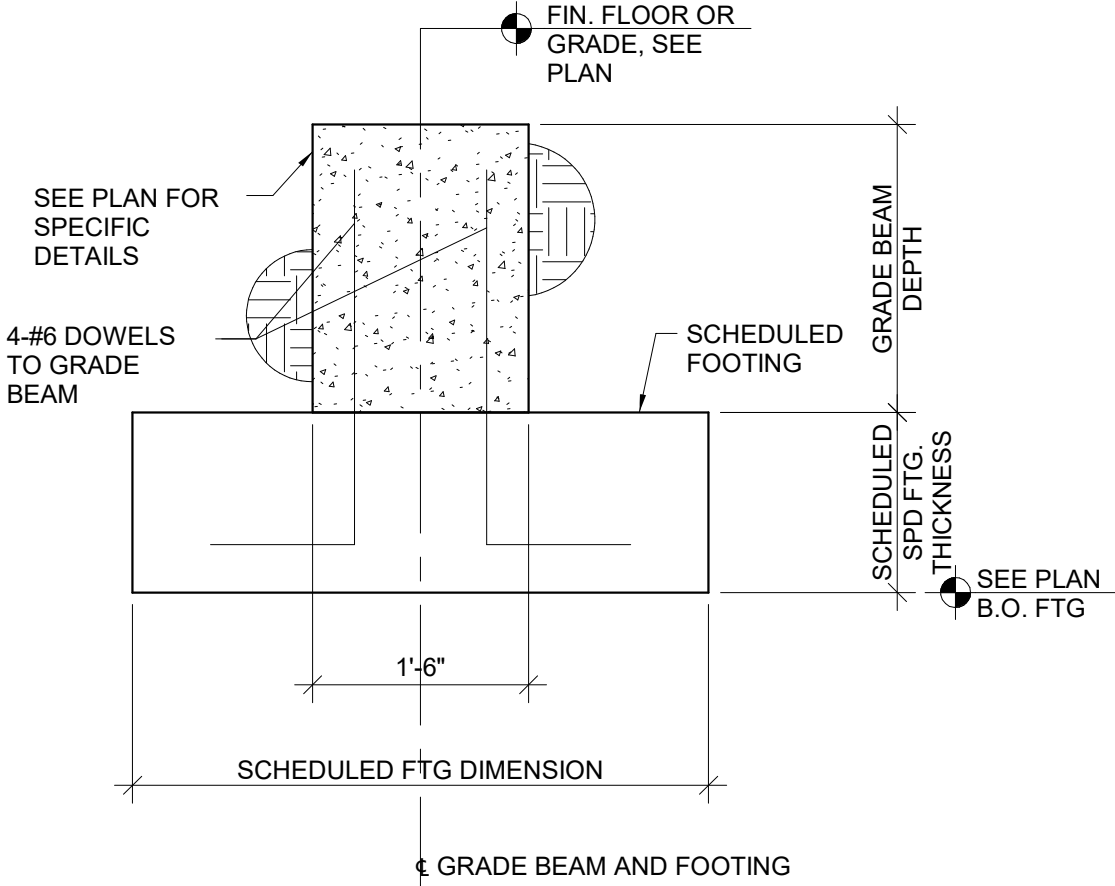
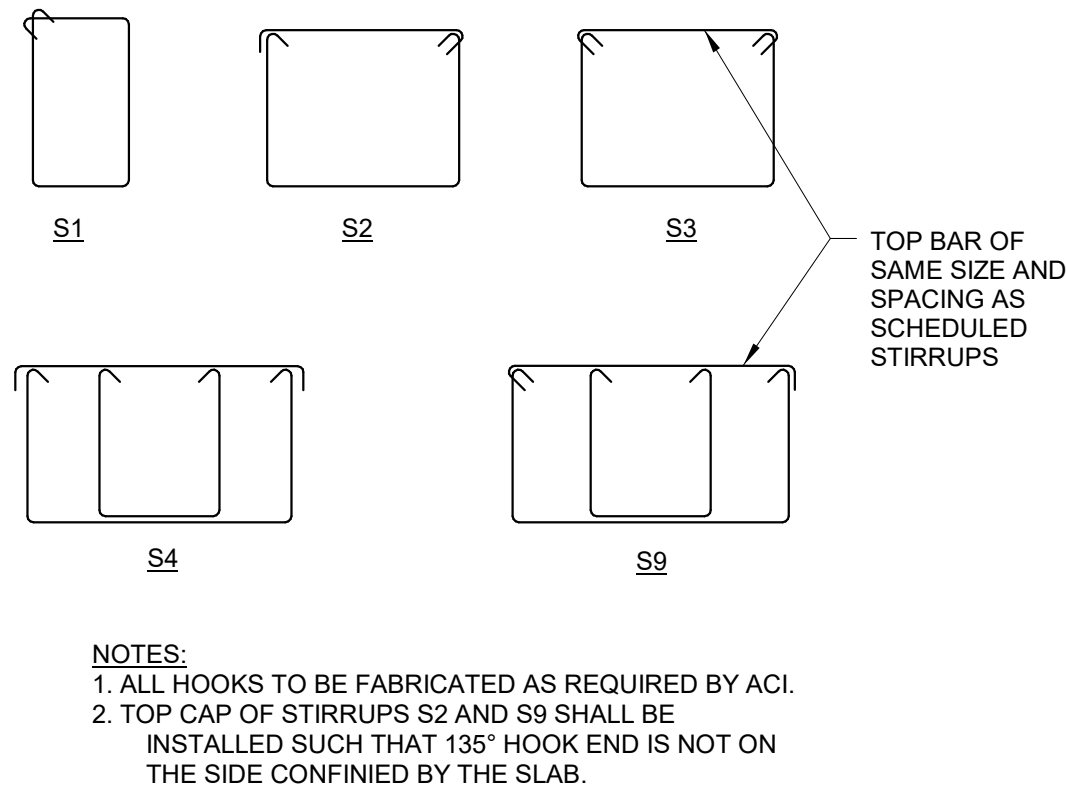
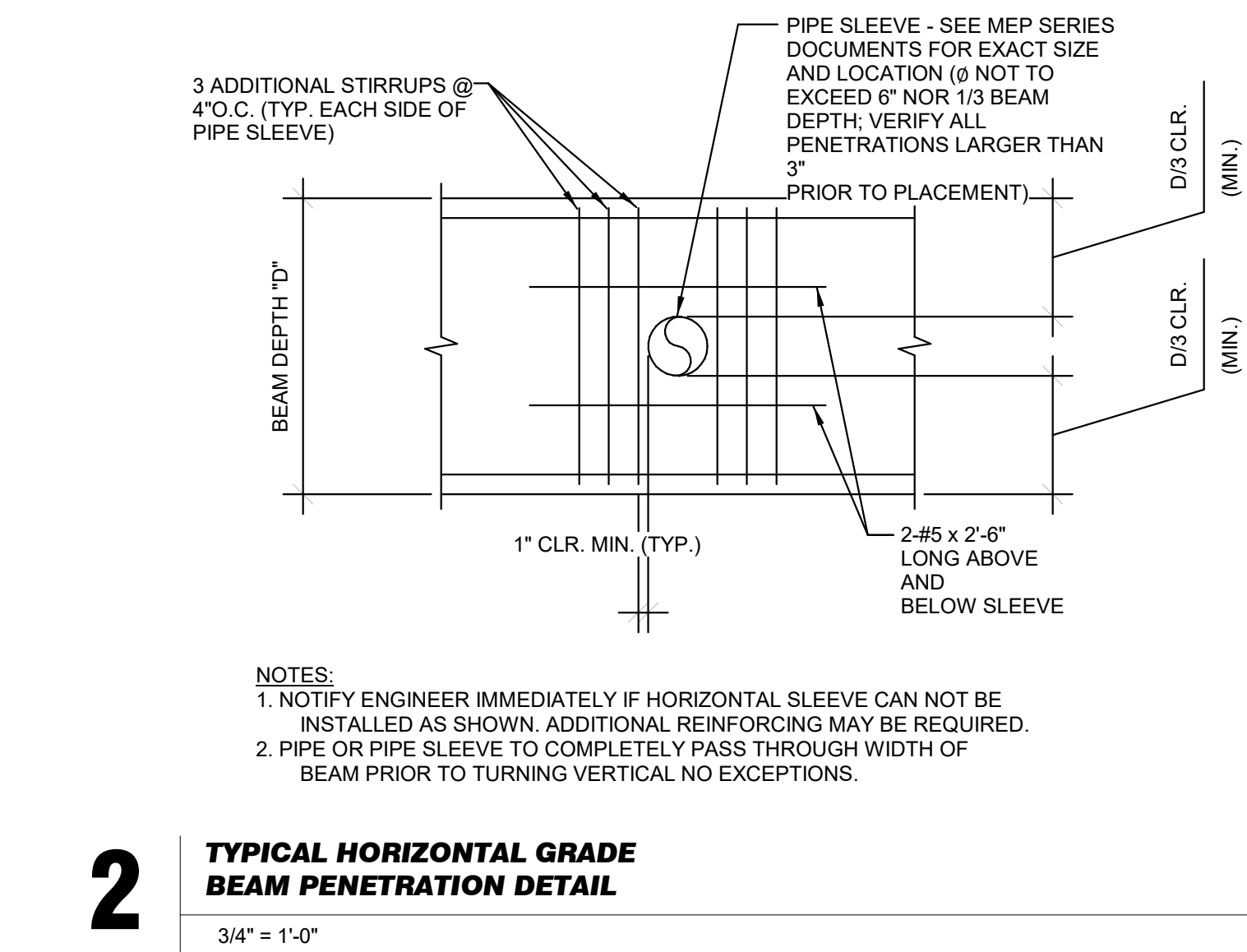




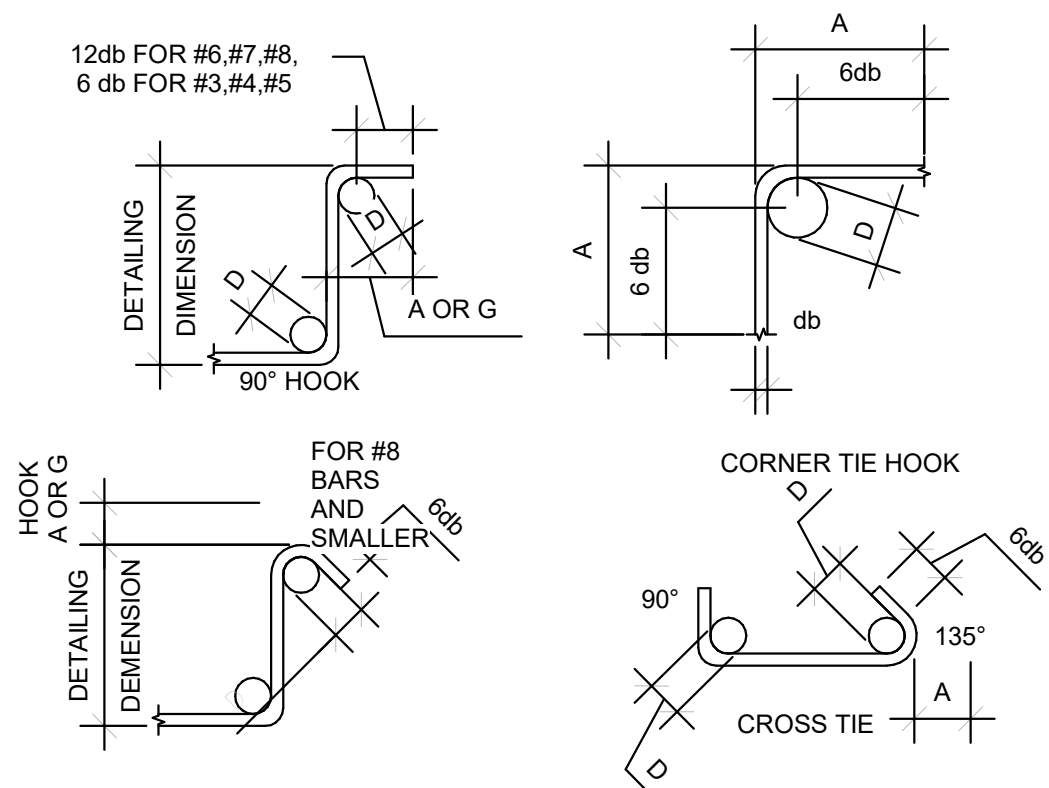
1 TYPICAL GRADE BEAM TO FOOTING DETAIL
3/4" = 1'-0"



5 TYPICAL STIRRUP TYPES
3/4" = 1'-0"



2 TYPICAL HORIZONTAL GRADE BEAM PENETRATION DETAIL
3/4" = 1'-0"

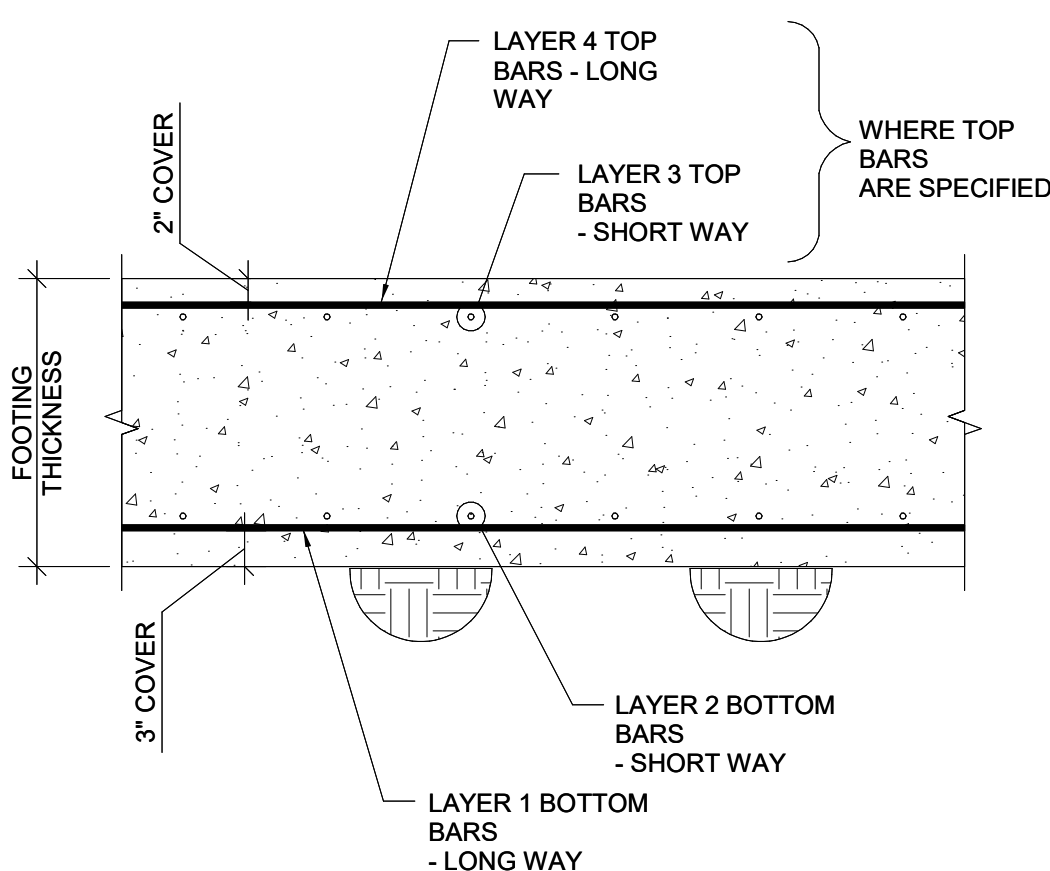


6 TYPICAL DETAIL STIRRUP AND TIE HOOK TYPES
3/4" = 1'-0"

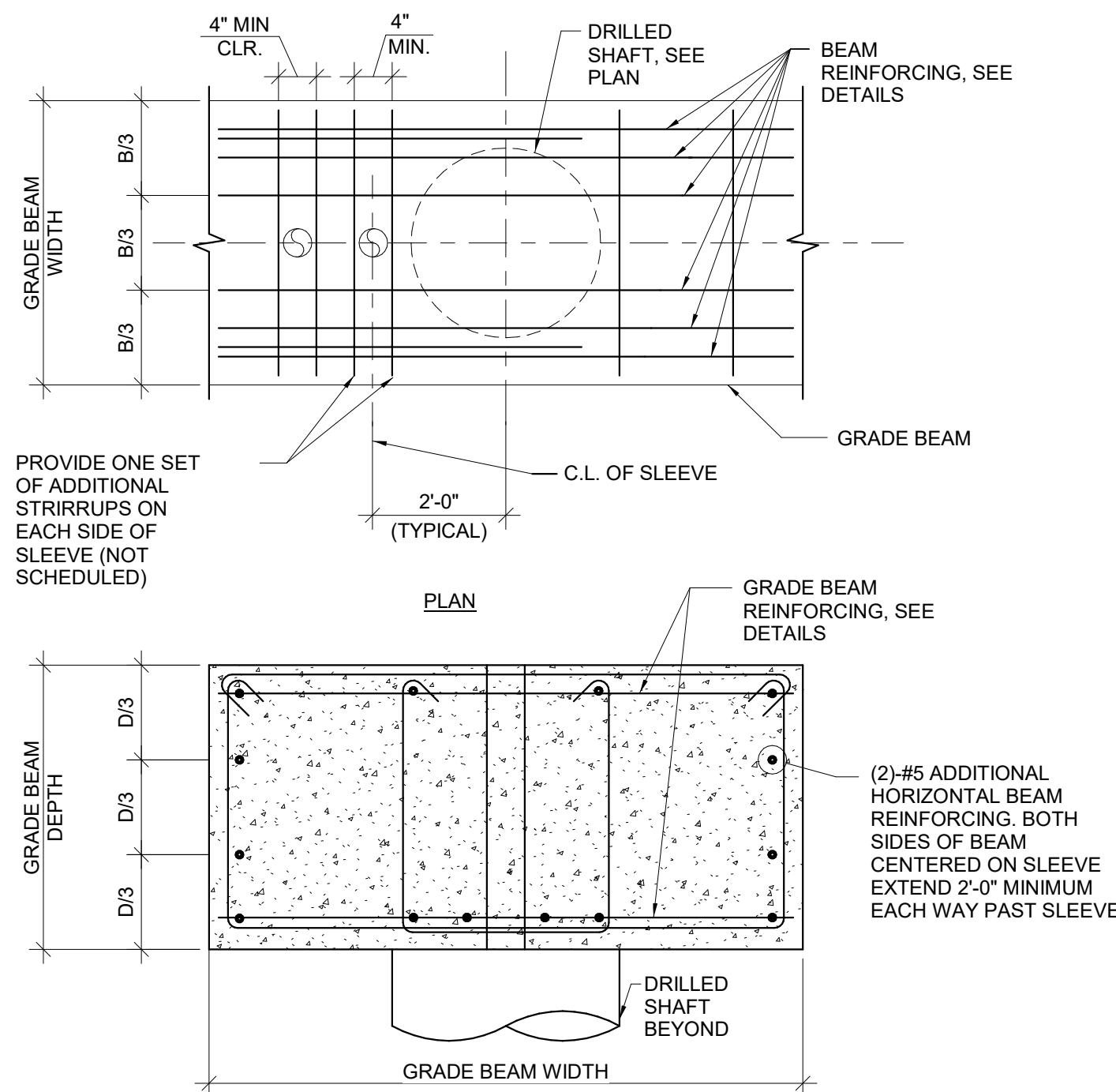
BAR SIZE	F'c=3000 psi		F'c=4000 psi		F'c=5000 psi	
	Id TOP	Id BOT	Id TOP	Id BOT	Id TOP	Id BOT
#3	2'-4"	1'-9"	2'-0"	1'-6"	1'-10"	1'-5"
#4	3'-1"	2'-4"	2'-8"	2'-1"	2'-5"	1'-10"
#5	3'-10"	3'-0"	3'-4"	2'-7"	3'-0"	2'-4"
#6	4'-8"	3'-7"	4'-0"	3'-1"	3'-7"	2'-9"
#7	6'-9"	5'-2"	5'-10"	4'-6"	5'-3"	4'-0"
#8	7'-9"	5'-11"	6'-8"	5'-2"	6'-0"	4'-7"
#9	8'-8"	6'-8"	7'-6"	5'-9"	6'-9"	5'-2"
#10	9'-10"	7'-6"	8'-6"	6'-6"	7'-7"	5'-10"
#11	10'-11"	8'-4"	9'-5"	7'-3"	8'-5"	6'-6"

NOTE:
FOR CLASS "A" SPLICE (PERMITTED ONLY WHEN NOT MORE THAN HALF THE BARS SPLICED & SPICES STAGGERED BY THE DISTANCE OF SPLICE LENGTH), USE SAME AS "Id" = TENSION DEVELOPMENT LENGTH TABLE.

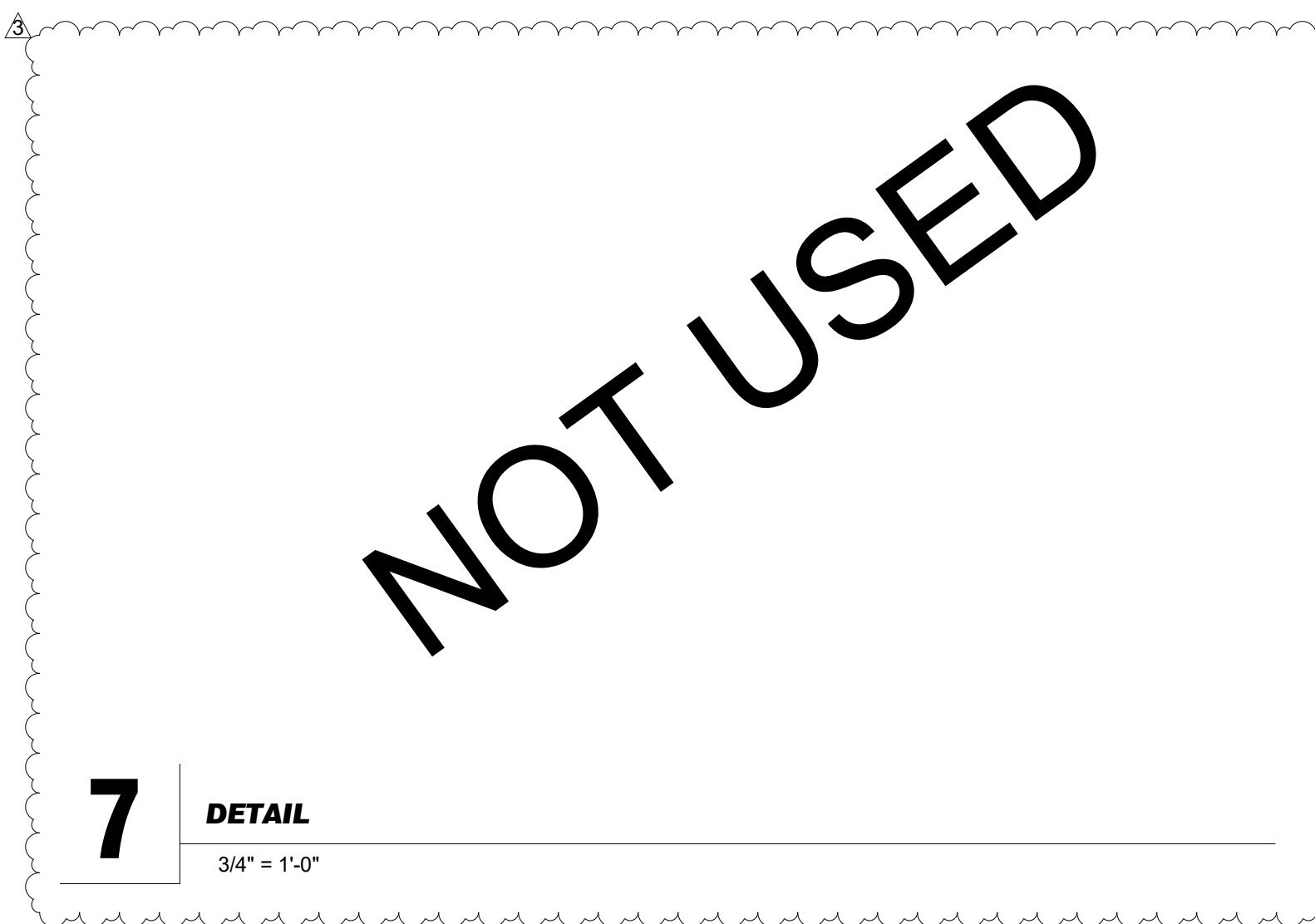
10 TENSION LAP SPLICES - CLASS B FOR TOP AND BOTTOM BARS
3/4" = 1'-0"



13 TYPICAL DETAIL BAR LAYER DIAGRAM RECTANGULAR SPREAD FOOTING
3/4" = 1'-0"



3 TYPICAL DETAIL VERTICAL PENETRATION, CONCRETE GRADE BEAM
3/4" = 1'-0"



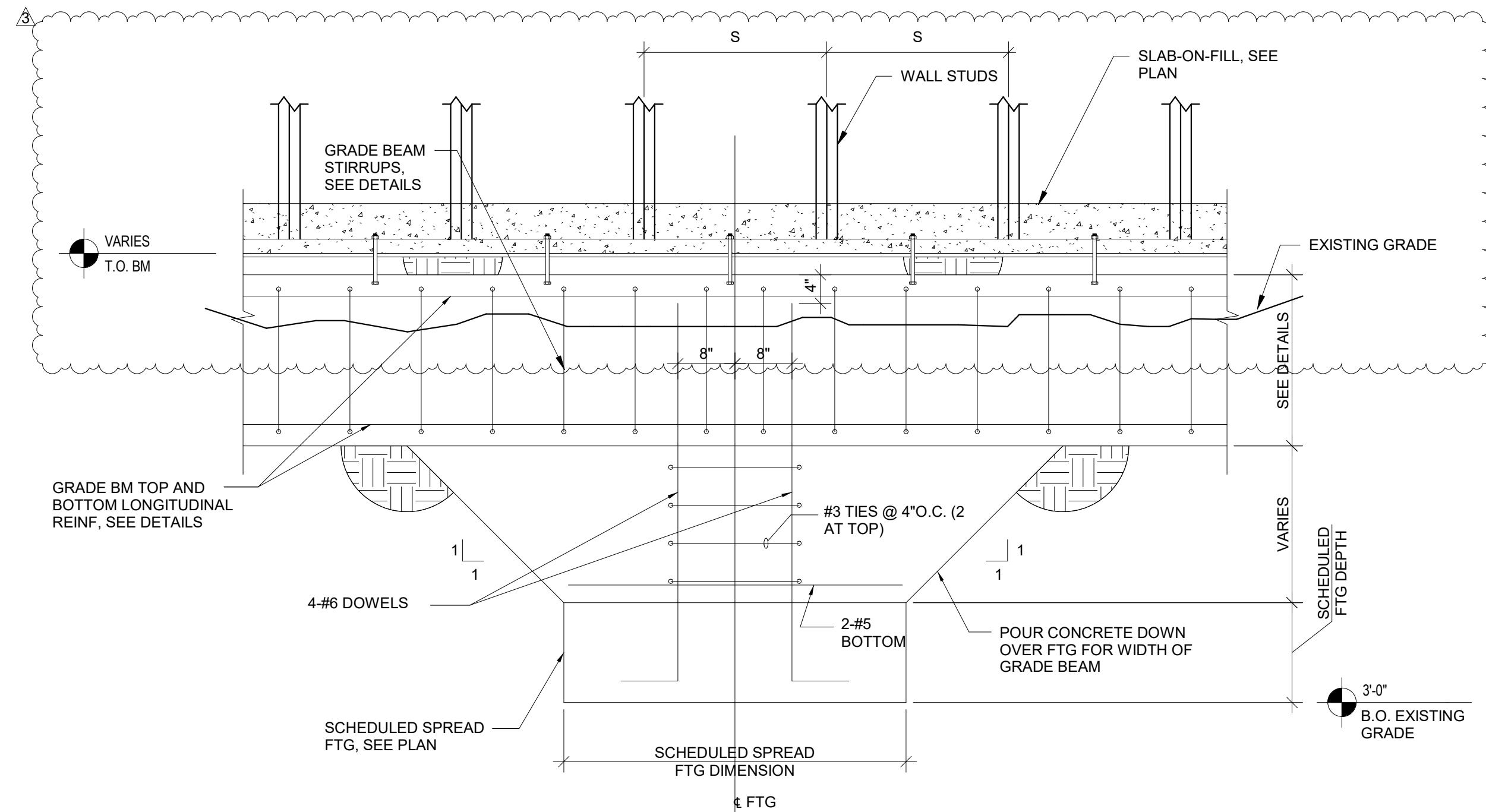
7 DETAIL
3/4" = 1'-0"

4 DETAIL
3/4" = 1'-0"

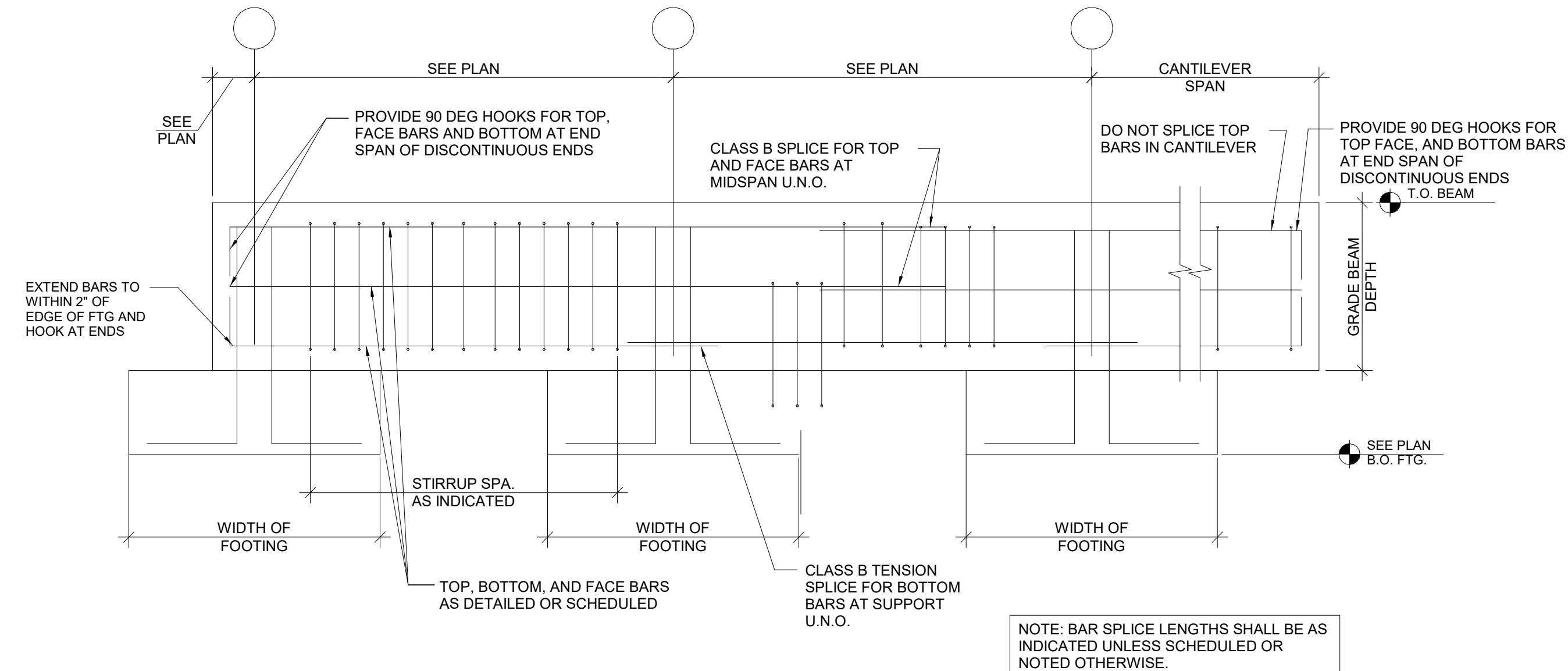
BAR SIZE	F'c=3000 psi		F'c=4000 psi		F'c=5000 psi	
	Id TOP	Id BOT	Id TOP	Id BOT	Id TOP	Id BOT
#3	1'-9"	1'-4"	1'-6"	1'-2"	1'-5"	1'-1"
#4	2'-4"	1'-10"	2'-1"	1'-7"	1'-10"	1'-5"
#5	3'-0"	2'-3"	2'-7"	2'-0"	2'-4"	1'-9"
#6	3'-7"	2'-9"	3'-1"	2'-4"	2'-9"	2'-1"
#7	5'-2"	4'-0"	4'-6"	3'-6"	4'-0"	3'-1"
#8	5'-11"	4'-7"	5'-2"	3'-11"	4'-7"	3'-6"
#9	6'-8"	5'-2"	5'-9"	4'-5"	5'-2"	4'-0"
#10	7'-6"	5'-10"	6'-6"	5'-0"	5'-10"	4'-6"
#11	8'-4"	6'-5"	7'-3"	5'-7"	6'-6"	5'-0"

NOTES:
1. "TOP" BARS ARE HORIZONTAL REBAR WITH MORE THAN 12 IN. OF FRESH CONCRETE CAST BELOW THE BARS AT THE DEVELOPMENT LENGTH.
2. "Id" FOR #3 & #4 BARS IN SLAB OR WALL ARE CONSERVATIVE & MAY BE REDUCED TO 0.75 TIMES (FOR #3 BARS) AND 0.94 TIMES (FOR #4 BARS) FROM THE TABULATED VALUES.
3. FOR LIGHT-WEIGHT CONCRETE MULTIPLY THE TABULATED VALUES BY 1.3.

8 TENSION DEVELOPMENT LENGTH FOR BEAM, SLAB AND WALL REBAR
3/4" = 1'-0"



11 TYPICAL DETAIL FOOTING BEARING DEPTH GREATER THAN BOTTOM OF GRADE BEAM
3/4" = 1'-0"



14 GRADE BEAM REINFORCING TYPICAL DETAIL
3/4" = 1'-0"

NOT USED

9 NOT USED
3/4" = 1'-0"

MARK	DIMENSIONS			REINFORCING	
	L	W	DEPTH	LONGITUDINAL	TRANSVERSE
F1	4'-0"	4'-0"	1'-4"	4-#6	4-#6
F2	5'-0"	5'-0"	1'-4"	5-#6	5-#6
F3	14'-10"	8'-0"	1'-6"	8-#6 T & B	14-#6 T & B

NOTES:
1. SCHEDULED REINFORCEMENT DENOTES BOTTOM BARS UNLESS NOTED OTHERWISE AS T&B; TOP AND BOTTOM.
2. SEE TYPICAL SPREAD FOOTING BAR DIAGRAM FOR LAYERED PLACEMENT OF BARS AS WELL AS DISTRIBUTION OF REINFORCING.

12 FOOTING SCHEDULE
3/4" = 1'-0"

CLIENT



ARCHITECT

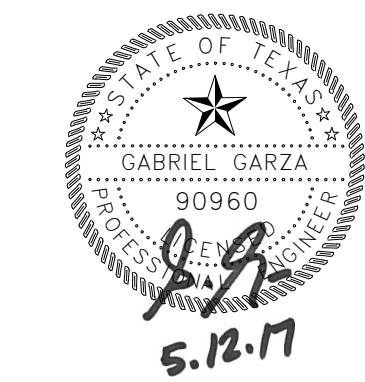


3131 BRIARPARK DR.
SUITE 200
HOUSTON, TX 77042
[T] 713 622 1444
[F] 713 968 9333
TBPE REG. No. F-2742
www.pgall.com

CONSULTANT

Garza & McLain
12313 Southwest Freeway, Suite 103
Sugar Land, Texas 77478
(281) 484-1234 (voice)
(281) 484-1234 (fax)

REGISTRATION



DRAWING HISTORY

NO.	ISSUE	DATE
1	ISSUE FOR BID	04/24/2017
2	ADDENDUM No. 1	04/27/2017
3	ADDENDUM No. 3	05/12/2017

PROJECT NAME
SAM HOUSTON VISITOR CENTER EXPANSION

PROJECT LOCATION
7600 TX-75
HUNTSVILLE, TX 77340

PROJECT NUMBER
1002602

SHEET TITLE
FOUNDATION DETAILS

SHEET NUMBER

S4.00

